

For each of the following two time-series regression models, and assuming MR1-MR6 hold, find $\text{var}(b_2|\mathbf{x})$ and examine whether the least squares estimator is consistent by checking whether $\lim_{T \rightarrow \infty} \text{var}(b_2|\mathbf{x}) = 0$.

- a. $y_t = \beta_1 + \beta_2 t + e_t, t = 1, 2, \dots, T$. Note that $\mathbf{x} = (1, 2, \dots, T)$, $\sum_{t=1}^T (t - \bar{t})^2 = \sum_{t=1}^T t^2 - \left(\sum_{t=1}^T t\right)^2 / T$, $\sum_{t=1}^T t = T(T+1)/2$ and $\sum_{t=1}^T t^2 = T(T+1)(2T+1)/6$.
- b. $y_t = \beta_1 + \beta_2 (0.5)^t + e_t, t = 1, 2, \dots, T$. Here, $\mathbf{x} = (0.5, 0.5^2, \dots, 0.5^T)$. Note that the sum of a geometric progression with first term r and common ratio r is

$$S = r + r^2 + r^3 + \dots + r^n = \frac{r(1 - r^n)}{1 - r}$$

- c. Provide an intuitive explanation for these results.

a.

$$\begin{aligned} \text{var}(b_2|\mathbf{x}) &= \frac{\sigma^2}{\sum_{t=1}^T (t - \bar{t})^2} \\ \sum_{t=1}^T (t - \bar{t})^2 &= \sum_{t=1}^T t^2 - \frac{(\sum_{t=1}^T t)^2}{T} \\ &= \frac{T(T+1)(2T+1)}{6} - \frac{\left(\frac{T(T+1)}{2}\right)^2}{T} \\ &= \frac{2T(T+1)(2T+1)}{12} - \frac{3T(T+1)^2}{12} \\ &= T(T+1) \frac{4T+2-3T-3}{12} \\ &= T(T+1) \frac{(T-1)}{12} = \frac{T(T^2-1)}{12} \\ \text{var}(b_2|\mathbf{x}) &= \frac{12\sigma^2}{T(T^2-1)} \end{aligned}$$

Then $\lim_{T \rightarrow \infty} \frac{12\sigma^2}{T(T^2-1)} = 0$, which means the least squares estimator b_2 is consistent.

b.

$$\begin{aligned} \sum_{t=1}^T (x_t - \bar{x})^2 &= \sum_{t=1}^T x_t^2 - \frac{(\sum_{t=1}^T x_t)^2}{T} \\ &= \frac{(1 - (\frac{1}{4})^T)}{3} - \frac{(1 - (\frac{1}{2})^T)^2}{T} \end{aligned} \quad \begin{cases} \sum x_t = \sum_{t=1}^T (\frac{1}{2})^t = \frac{\frac{1}{2}(1 - (\frac{1}{2})^T)}{1 - \frac{1}{2}} = 1 - (\frac{1}{2})^T \\ \sum x_t^2 = \sum_{t=1}^T (\frac{1}{4})^t = \frac{\frac{1}{4}(1 - (\frac{1}{4})^T)}{1 - \frac{1}{4}} = \frac{(1 - (\frac{1}{4})^T)}{3} \end{cases}$$

$$\text{var}(b_2|\mathbf{x}) = \frac{\sigma^2}{\frac{(1 - (\frac{1}{4})^T)}{3} - \frac{(1 - (\frac{1}{2})^T)^2}{T}}$$

$\lim_{T \rightarrow \infty} \text{var}(b_2|\mathbf{x}) = 3\sigma^2 \neq 0$, which means the least squares estimator b_2 is not consistent.

c.

In (a), the x_t 's become more dispersed as T increases, so OLS learns the slope precisely.
In (b), the x_t 's quickly bunch near 0, the extra data add little information, and OLS cannot estimate the slope consistently.

Suppose that, from a sample of 63 observations, the least squares estimates and the corresponding estimated covariance matrix are given by

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} \quad \widehat{\text{cov}}(b_1, b_2, b_3) = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

Using a 5% significance level, and an alternative hypothesis that the equality does not hold, test each of the following null hypotheses:

- a. $\beta_2 = 0$
- b. $\beta_1 + 2\beta_2 = 5$
- c. $\beta_1 - \beta_2 + \beta_3 = 4$

a. $H_0: \beta_2 = 0$
 $\text{var}(b_2) = 4$
 $b_2 = 3$
 $t = \frac{3-0}{2} = 1.5$

b. $H_0: \beta_1 + 2\beta_2 = 5$
 $\text{var}(b_1 + 2b_2) = \text{var}(b_1) + 4\text{var}(b_2) + 4\text{cov}(b_1, b_2)$
 $= 3 + 4 \cdot 4 + 4 \cdot (-2)$
 $= 11$

$df = 63 - 3 = 60$

$\text{se}(b_1 + 2b_2) = \sqrt{11}$

$t_{(0.975, 60)} \approx 2$

$t = \frac{8-5}{\sqrt{11}} \approx 0.9$

$\therefore 1.5 < 2$

$t_{(0.975, 60)} \approx 2$

$\therefore$ Fail to reject H_0

$\therefore 0.9 < 2$

$\therefore$ Fail to reject H_0

c.

$\beta_1 - \beta_2 + \beta_3 = 4$

$\text{var}(b_1 - b_2 + b_3) = \text{var}(b_1) + \text{var}(b_2) + \text{var}(b_3) - 2\text{cov}(b_1, b_2) + 2\text{cov}(b_1, b_3) - 2\text{cov}(b_2, b_3)$
 $= 3 + 4 + 3 + 4 + 1$
 $= 15$

$\text{se}(b_1 - b_2 + b_3) = \sqrt{15}$

$t = \frac{-2-4}{\sqrt{15}} \approx -1.55$

$t_{0.975, 60} \approx 2$

$\therefore 1.55 < 2$

$\therefore$ Fail to reject H_0

A generalized version of the estimator for β_2 proposed in Exercise 2.9 by Professor I.M. Mean for the regression model $y_i = \beta_1 + \beta_2 x_i + e_i, i = 1, 2, \dots, N$ is

$$\hat{\beta}_{2,mean} = \frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1}$$

where $(\bar{y}_1, \bar{x}_1)$ and $(\bar{y}_2, \bar{x}_2)$ are the sample means for the first and second halves of the sample observations, respectively, after ordering the observations according to increasing values of x . Given that assumptions MR1-MR6 hold:

- Show that $\hat{\beta}_{2,mean}$ is unbiased.
- Derive an expression for $\text{var}(\hat{\beta}_{2,mean} | \mathbf{x})$.
- Write down an expression for $\text{var}(\hat{\beta}_{2,mean})$.
- Under what conditions will $\hat{\beta}_{2,mean}$ be a consistent estimator for β_2 ?
- Randomly generate observations on x from a uniform distribution on the interval (0,10) for sample sizes $N = 100, 500, 1000$, and, if your software permits, $N = 5000$. Assuming $\sigma^2 = 1000$, for each sample size, compute:
 - $\text{var}(b_2 | \mathbf{x})$ and $\text{var}(\hat{\beta}_{2,mean} | \mathbf{x})$ where b_2 is the OLS estimator.
 - Estimates for $E[(s_x^2)^{-1}]$ and $E[4/(\bar{x}_2 - \bar{x}_1)^2]$ where s_x^2 is the sample standard deviation for x using N as a divisor.
- Comment on the relative magnitudes of your answers in part (e), (i) and (ii) and how they change as sample size increases. Does it appear that $\hat{\beta}_{2,mean}$ is consistent?
- Show that $E[(s_x^2)^{-1}] \xrightarrow{P} 0.12$ and $E[4/(\bar{x}_2 - \bar{x}_1)^2] \xrightarrow{P} 0.16$. [Hint: The variance of a uniform random variable defined on the interval (a, b) is $(b-a)^2/12$.]
- Suppose that the observations on x were not ordered according to increasing magnitude but were randomly assigned to any position. Would the estimator $\hat{\beta}_{2,mean}$ be consistent? Why or why not?

(a.) NTS: $\hat{\beta}_{2,mean}$ is unbiased.

$$\bar{y}_1 = \beta_1 + \beta_2 \bar{x}_1 + \bar{e}_1$$

$$\bar{y}_2 = \beta_1 + \beta_2 \bar{x}_2 + \bar{e}_2$$

$$\bar{y}_2 - \bar{y}_1 = \beta_2 (\bar{x}_2 - \bar{x}_1) + (\bar{e}_2 - \bar{e}_1)$$

$$\hat{\beta}_{2,mean} = \frac{\beta_2 (\bar{x}_2 - \bar{x}_1) + (\bar{e}_2 - \bar{e}_1)}{\bar{x}_2 - \bar{x}_1} = \beta_2 + \frac{(\bar{e}_2 - \bar{e}_1)}{\bar{x}_2 - \bar{x}_1}$$

(c.)

$$\begin{aligned} \text{var}(\hat{\beta}_{2,mean}) &= E(\text{var}(\hat{\beta}_{2,mean} | \mathbf{x})) + \text{var}(E(\hat{\beta}_{2,mean} | \mathbf{x})) \\ &= E\left[\frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}\right] + 0 \\ &= \frac{4\sigma^2}{N} E\left[\frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\right] \end{aligned}$$

By MR4, $E(e_i | x_i) = 0$, then $E(\bar{e}_2 - \bar{e}_1 | \mathbf{x}) = 0$

$\therefore E(\hat{\beta}_{2,mean} | \mathbf{x}) = \beta_2$, which is unbiased.

(b.)

$$\text{var}(\hat{\beta}_{2,mean} | \mathbf{x}) = \text{var}\left(\frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1} | \mathbf{x}\right) = \frac{1}{(\bar{x}_2 - \bar{x}_1)^2} \text{var}(\bar{e}_2 - \bar{e}_1 | \mathbf{x}) = \frac{1}{(\bar{x}_2 - \bar{x}_1)^2} \cdot \frac{4\sigma^2}{N} = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}$$

$$\begin{aligned} \text{By MR5, } \text{var}(e_i | \mathbf{x}) &= \sigma^2 \\ \text{Then } \text{var}(\bar{e}_1 | \mathbf{x}) &= \text{var}\left(\frac{1}{N} \sum e_i | \mathbf{x}\right) = \left(\frac{1}{N}\right)^2 \cdot \left(\frac{N}{2}\right) \cdot \sigma^2 = \frac{\sigma^2}{2N} \\ \text{var}(\bar{e}_2 - \bar{e}_1 | \mathbf{x}) &= \text{var}(\bar{e}_1 | \mathbf{x}) + \text{var}(\bar{e}_2 | \mathbf{x}) = \frac{\sigma^2}{2N} + \frac{\sigma^2}{2N} = \frac{4\sigma^2}{N} \end{aligned}$$

(d.)

NTS: $\hat{\beta}_{2,mean} \xrightarrow{P} \beta_2$, when $N \rightarrow \infty$

if $\text{var}(\hat{\beta}_{2,mean}) \rightarrow 0$, then $\frac{1}{N} E\left[\frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\right] \rightarrow 0$ — (1)

if $\bar{x}_2 - \bar{x}_1 \xrightarrow{P} c$, where $c \neq 0$, then $E\left[\frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\right] \rightarrow \frac{1}{c^2}$ — (2)

$\therefore$ By (1) & (2), under $\bar{x}_2 - \bar{x}_1 \xrightarrow{P} c$, where $c \neq 0$, $\frac{1}{N} E\left[\frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\right] \rightarrow 0$ as $N \rightarrow \infty$

$\therefore \text{var}(\hat{\beta}_{2,mean}) \rightarrow 0$, then $\hat{\beta}_{2,mean} \xrightarrow{P} \beta_2$

(e.)

N	$\text{var}(b_2 \mathbf{x})$	$\text{var}(\hat{\beta}_{2,mean} \mathbf{x})$	$E[(s_x^2)^{-1}]$	$E[4/(\bar{x}_2 - \bar{x}_1)^2]$
100	1.2210	1.6473	0.1221	0.1647
500	0.2410	0.3222	0.1205	0.1612
1000	0.1201	0.1604	0.1202	0.1604
5000	0.0240	0.0320	0.1200	0.1601

(f.)

As N increases, both $\text{var}(b_2 | \mathbf{x})$ & $\text{var}(\hat{\beta}_{2,mean} | \mathbf{x})$ decrease approximately $1/N$.

However, $\text{var}(\hat{\beta}_{2,mean} | \mathbf{x})$ is more larger, because $E[4/(\bar{x}_2 - \bar{x}_1)^2] \rightarrow 0.16$ & $E[(s_x^2)^{-1}] \rightarrow 0.12$

Hence, for larger N , $E(\text{var}(\hat{\beta}_{2,mean} | \mathbf{x})) \approx \frac{(0.16)\sigma^2}{N} > E[\text{var}(b_2 | \mathbf{x})] \approx \frac{0.12\sigma^2}{N}$

Thus, the OLS is more efficient, but both $\rightarrow 0$ as $N \rightarrow \infty$. Therefore, it appears $\hat{\beta}_{2,mean}$ is consistent.

g. Show that $E[(S_x^2)^{-1}] \xrightarrow{P} 0.12$ and $E[4/(\bar{x}_2 - \bar{x}_1)^2] \xrightarrow{P} 0.16$. [Hint: The variance of a uniform random variable defined on the interval (a, b) is $(b - a)^2/12$.]

h. Suppose that the observations on x were not ordered according to increasing magnitude but were randomly assigned to any position. Would the estimator $\hat{\beta}_{2, \text{mean}}$ be consistent? Why or why not?

(g.) NTS: $E[(S_x^2)^{-1}] \xrightarrow{P} 0.12$
 $\because X \sim U(0, 10)$
 $\therefore \text{Var}(X) = \frac{(b-a)^2}{12} = \frac{25}{3}$
 $\therefore S_x^2 = \frac{1}{N} \sum_{i=1}^N (X_i - \bar{X})^2$
 $\therefore S_x^2 \xrightarrow{P} \text{Var}(X)$
 $\therefore (S_x^2)^{-1} \xrightarrow{P} \left(\frac{25}{3}\right)^{-1}$
 $\therefore E[(S_x^2)^{-1}] \rightarrow \frac{12}{100} = 0.12$

NTS: $E[4/(\bar{x}_2 - \bar{x}_1)^2] \xrightarrow{P} 0.16$
 $\because \bar{x}_2 \xrightarrow{P} 7.5, \bar{x}_1 \xrightarrow{P} 2.5$
 $\therefore \bar{x}_2 - \bar{x}_1 \xrightarrow{P} 5$
 $\therefore \frac{4}{(\bar{x}_2 - \bar{x}_1)^2} \xrightarrow{P} \frac{4}{25}$
 $\therefore E[4/(\bar{x}_2 - \bar{x}_1)^2] \xrightarrow{P} 0.16$

(h.)
 No, it will generally not be consistent.
 Because if $\bar{x}_2 - \bar{x}_1 \rightarrow 0$, $\hat{\beta}_{2, \text{mean}}$ will be unstable, which means that $\hat{\beta}_{2, \text{mean}} \not\rightarrow \beta_2$